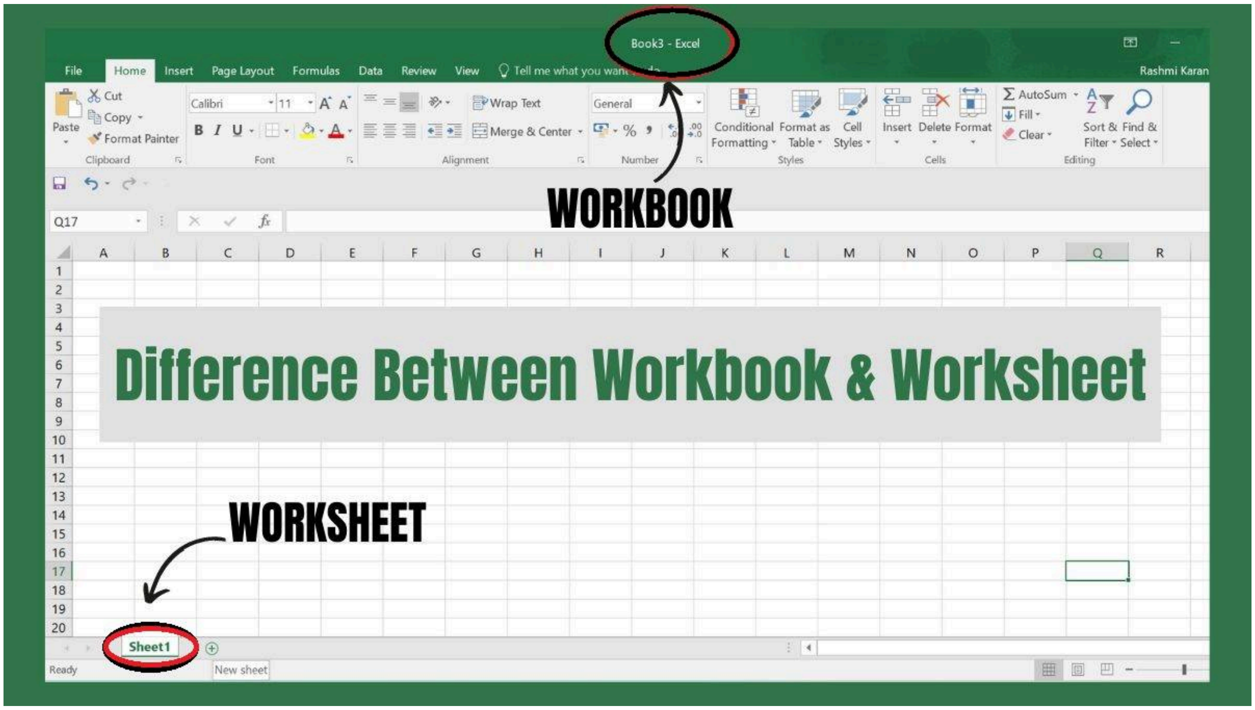
Chap 1: Introduction to Microsoft Excel

Concepts of Work book & Worksheets

The difference between a workbook and a worksheet must confuse you. So, are workbooks and worksheets the same? Not quite! Both play a distinct role in organising and analysing your data. Workbooks provide the overall structure and big-picture view, while worksheets offer focused analysis and manipulation.



Tabular Comparison - Workbook vs Worksheet

The main difference between workbook and worksheet is that a workbook comprises multiple worksheets, providing a structured framework to organise and manage multiple data sets, while a worksheet is a single "page" within a workbook where users can focus on a specific dataset.

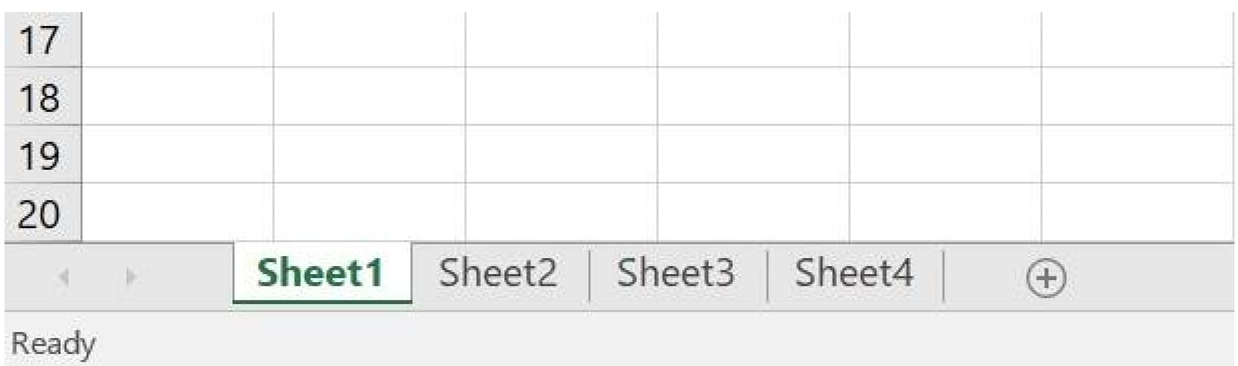
Feature	Workbook	Worksheet

Definition	A container file that holds multiple worksheets.	A single sheet within a workbook that holds data and formatting.
Purpose	Organizes and manages related sets of data.	Presents and manipulates specific data sets.
Number per file	One per file.	Multiple per workbook (limited by memory).
Data storage	Stores all data for all worksheets in the workbook.	Stores data specific to that worksheet.
Formatting	Can have global formatting settings that apply to all worksheets.	Individual formatting can be applied to cells, rows, and columns.
Collaborati on	Multiple users can work on different worksheets in the same workbook simultaneously.	Collaboration features are limited to the specific worksheet.
Examples	Financial reports with multiple sheets for income, expenses, and budgets.	A product list with columns for product name, price, and quantity.

Complexity	It can be highly complex with multiple sheets, formulas across sheets, and macros.	Usually simpler, focused on specific data sets within a single sheet.
Security	Can implement password protection and sheet-level permissions for better security.	Security options are limited to the worksheet.
Data Validation	Global data validation rules can be applied across all worksheets.	Data validation rules are specific to the current worksheet.

What is an Excel Worksheet?

An Excel worksheet is a single matrix arrangement composed of rows and columns, starting from row 1 and column A. Consider a worksheet as a digital canvas where users can organise, store, and manipulate data, texts, and formulas. Each worksheet is included within a workbook and can be accessed by clicking on its corresponding tab at the bottom of the Excel window



Spreadsheets in Excel can accommodate various types of information, including numerical data, formulas, tables, and graphical representations.

Features of an Excel Worksheet

Here are some key features of an Excel worksheet:

- **Cell-Based Data Entry:** Users can enter and manipulate data in individual cells, each identified by a unique column letter and row number.
- **Formula Bar:** The formula bar allows users to view and edit the contents of the active cell, including entering formulas and functions.
- **Rows and Columns:** Worksheets are organized into rows (numbered) and columns (lettered), providing a grid structure for organizing and displaying data.
- **AutoFill:** Excel's AutoFill feature allows users to quickly fill cells with sequential data or data patterns, such as numbers, dates, or custom lists, by dragging the fill handle.
- **Cell References**: Users can create formulas referencing other cells within the same worksheet or across different worksheets within the same workbook.
- **Charts and Graphs:** Worksheets support the creation of various charts and graphs to visualize data trends and patterns, enhancing data analysis and presentation.
- **Worksheet Protection**: Users can protect worksheets by password, restricting editing, formatting, or structural changes to prevent unauthorized modifications.

What is an Excel Workbook?

An Excel workbook is an Excel file that can contain one or multiple spreadsheets. By default, a new workbook provides three worksheets. Workbooks serve as containers for organizing related information or datasets.

Workbooks can collect separate or interrelated data and are saved with file extensions such as .xls or .xlsx. They offer the flexibility to manage and manipulate multiple data sets within a single file, facilitating organization and accessibility.

Features of an Excel Workbook

Here are some key features of an Excel workbook:

- **Includes Multiple Worksheets:** Workbooks can contain multiple worksheets, where users can organize and manage different data sets.

- **Formatting Tools:** A workbook contains a wide range of formatting options, including font styles, colours, borders, and cell alignment, to customise the appearance of data.
- **Formulas and Functions:** Users can perform calculations and data analysis using built-in formulas and functions, such as SUM, AVERAGE, IF, VLOOKUP, etc.
- **Charts and Graphs:** Users can create charts and graphs to visualize data trends and patterns.
- **PivotTables:** PivotTables allow users to summarize, analyze, and present data more dynamically and in a more in-depth manner.
- **Data Import and Export:** You can import data from external sources such as databases, text files, and web queries. Excel also allows data export to various file formats for sharing and collaboration.
- **Collaboration Tools:** Excel offers collaboration features such as shared workbooks, comments, and track changes, allowing multiple users to work on the same workbook simultaneously.

Various Data Types:

Number data

Number data can be any value, including large numbers, small fractions, or qualitative data. In this section, we will take a look at examples including currency amounts, whole numbers, percentages, decimals, dates, times, and telephone numbers. As we will see, in order to ensure that Excel interprets your numeric data accurately, we have to define them using proper symbols and formats.

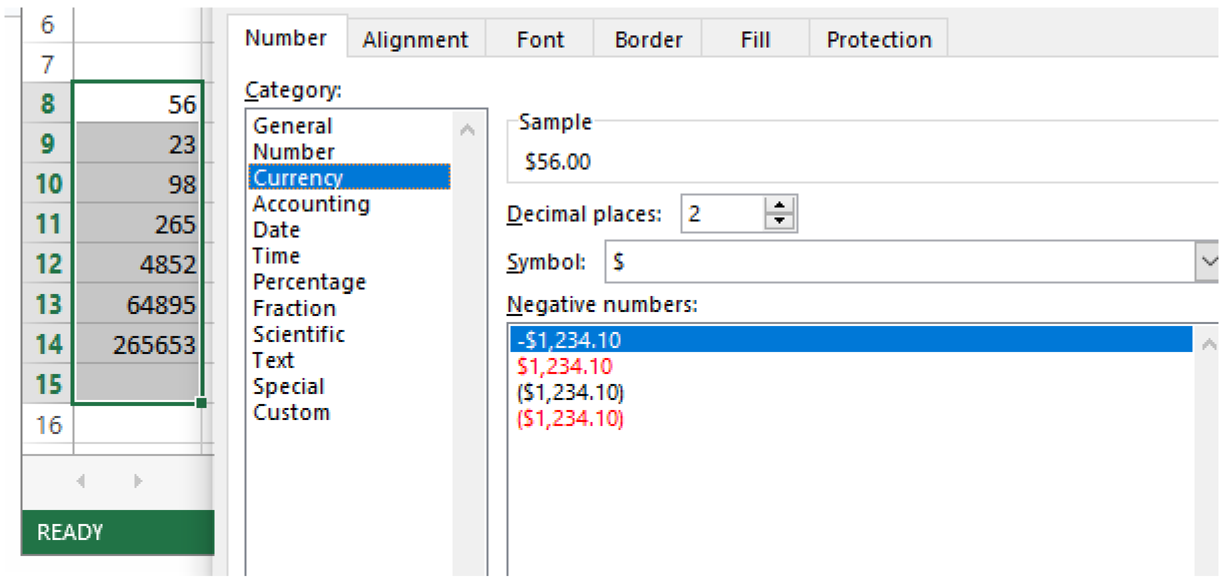
Keep in mind, there are subtle differences. For example, in a case where one cell has financial data and another has a date, Excel registers them both as Numeric, yet they are not identical.

Currency

You will be familiar with the Currency data type if you work with financial data. It formats monetary values and ensures that financial data is accurately represented by appropriate currency symbols and decimal places.

Here’s how you can apply the currency data type. Here we will convert numbers into currency.

- Select the range of cells you want to change.
- Right-click.
- Select the Format cell option, and a dialog box will appear.
- Go to the Number tab.
- Select the Currency data type and format your values.



Date and Time

Date and Time data types store dates and times in different formats. These formats help with chronological data analysis, scheduling, and time-sensitive calculations.

Let's take a look at the example of how to convert a Text into a Date. For time formatting, follow these steps and select the Time data type instead.

- First, select the cell you want to change.
- Right-click on the cell.
- Select the Format cells option.

- Go to the Number tab.
- Select the Date data type and format your values.

Depending on your goal, there might be additional steps:.

- If you want to write the current time, simply write =NOW().
- To convert the date to Text, all you have to do is apply =TEXT(Cell number, "MM/DD/YYYY").

Percentage

The Percentage data type converts numbers into percentages, making it easier to read and interpret ratio data and proportional values.

Fractions

You can use the Fraction data type to display your value in fractions instead of decimals.

Scientific

The Scientific data type displays a number in exponential notation, using E+n to represent that the number preceding E is multiplied by 10 to the nth power.

Special

Excel's Special data type includes formatting for zip codes, phone numbers, and Social Security numbers (SSNs). These formats keep the leading zeros in the case of zip codes and correctly format phone numbers and SSNs using the appropriate separators.

Custom

As the name suggests, Custom data type allows you to customize the formatting according to your needs. It provides various formatting suggestions that can help you customize your values.

Text data

Text data is the basic type that allows you to input characters, including alphabetical, numerical, and special symbols.

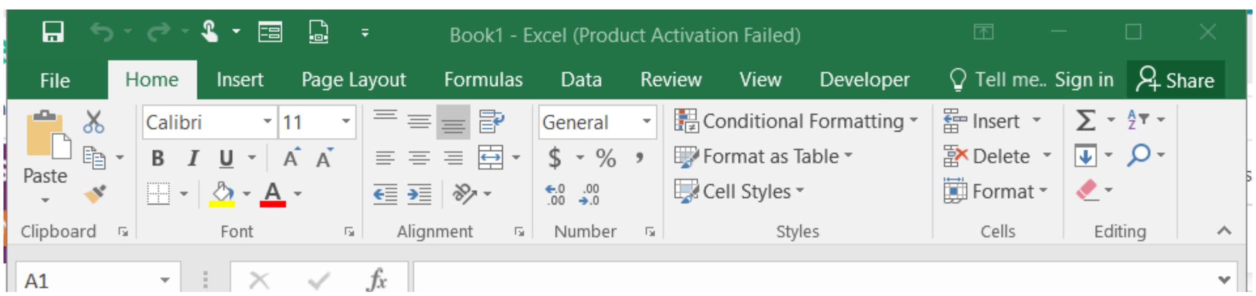
Boolean data

Excel's Boolean data type represents logical values that perform logical operations. It only has two possible values: TRUE and FALSE. These values are used in functions and formulas to test conditions and return logical results.

Features of MS Excel

Ribbon

Th eRibbon in MS-Excel is the topmost row of tabs that provide the user with different facilities/functionalities. These tabs are:



Home Tab

It provides the basic facilities like changing the font, size of text, editing the cells in the spreadsheet, autosum, etc.

Insert Tab

It provides the facilities like inserting tables, pivot tables, images, clip art, charts, links, etc.

Page layout

It provides all the facilities related to the spreadsheet-like margins, orientation, height, width, background etc. The worksheet appearance will be the same in the hard copy as well.

Formulas

It is a package of different in-built formulas/functions which can be used by user just by selecting the cell or range of cells for values.

Data

The Data Tab helps to perform different operations on a vast set of data like analysis through what-if analysis tools and many other data analysis tools, removing duplicate data, transpose the row and column, etc. It also helps to access data(s) from different sources as well, such as from Ms-Access, from web, etc.

Review

This tab provides the facility of thesaurus, checking spellings, translating the text, and helps to protect and share the worksheet and workbook.

View

It contains the commands to manage the view of the workbook, show/hide ruler, gridlines, etc, freezing panes, and adding macros.

How to Create a New Spreadsheet

In Excel 3 sheets are already opened by default, now to add a new sheet :

- In the lowermost pane in Excel, you can find a button.
- Click on that button to add a new sheet.
- We can also achieve the same by Right-clicking on the sheet number before which you want to insert the sheet.

- Click on Insert.Select Worksheet.
- Click OK.

. Inserting Columns and Rows

- **Inserting Columns:**
  - **Excel/Google Sheets:**
    1. Right-click the column header where you want to insert a new column. For example, if you want to insert a column before column B, right-click the header "B".
    2. Select **Insert** from the context menu. This will insert a new column before the selected column.
    3. Alternatively, you can select a column, go to the **Home** tab, and click **Insert** in the Cells group.
- **Inserting Rows:**
  - **Excel/Google Sheets:**
    1. Right-click the row number where you want to insert a new row. For example, if you want to insert a row above row 3, right-click the number "3".
    2. Select **Insert** from the context menu. This will insert a new row above the selected row.
    3. Alternatively, you can select a row, go to the **Home** tab, and click **Insert** in the Cells group.

- **Multiple Insertions:**
  - If you want to insert multiple rows or columns, select the number of rows or columns you need. Then, right-click and choose **Insert**, or use the **Insert** option from the toolbar. The software will insert the same number of rows or columns as you selected.

2. Removing Columns and Rows

- **Removing Columns:**
  - **Excel/Google Sheets:**
    1. Right-click the column header you want to delete. For example, if you want to delete column B, right-click the header "B".
    2. Choose **Delete** from the context menu. This will remove the entire column.
    3. Alternatively, you can select the column and press **Ctrl + - (minus)** on your keyboard (Windows) or **Cmd + - (minus)** (Mac).
- **Removing Rows:**
  - **Excel/Google Sheets:**
    1. Right-click the row number you want to delete. For example, if you want to delete row 3, right-click the number "3".
    2. Choose **Delete** from the context menu. This will remove the entire row.
    3. Alternatively, you can select the row and press **Ctrl + - (minus)** on your keyboard (Windows) or **Cmd + - (minus)** (Mac).
- **Multiple Deletions:**

- To delete multiple rows or columns, select the rows or columns you want to remove, right-click, and select **Delete**.

3. Resizing Columns and Rows

- **Resizing Columns:**
  - **Excel/Google Sheets:**
    1. Hover your cursor between two column headers (for example, between columns A and B) until the cursor changes to a double-sided arrow.
    2. Click and drag the boundary to resize the column.
    3. Alternatively, double-click the boundary between columns to auto-resize the column to fit the contents.
- **Resizing Rows:**
  - **Excel/Google Sheets:**
    1. Hover your cursor between two row numbers (for example, between rows 2 and 3) until the cursor changes to a double-sided arrow.
    2. Click and drag the boundary to resize the row.
    3. Alternatively, double-click the boundary between rows to auto-resize the row to fit the contents.
- **Multiple Resizing:**
  - To resize multiple rows or columns at once, select the rows or columns you want to resize. Then, drag the boundary between any two selected rows or columns, and all of them will resize together.

Tips for Efficient Row and Column Management

- **AutoFit:**
  - If you want columns or rows to automatically adjust to the size of the content, double-click on the boundary between headers (columns or rows). This feature is called **AutoFit** and will ensure that the column or row is resized based on the largest content in that column or row.
- **Keyboard Shortcuts:**
  - **Insert Column:**
    - **Excel (Windows):** Alt + H, then I + C
    - **Excel (Mac):** Cmd + Shift + K
  - **Insert Row:**
    - **Excel (Windows):** Alt + H, then I + R
    - **Excel (Mac):** Cmd + Shift + R
  - **Delete Row/Column:**
    - **Windows:** Ctrl + -
    - **Mac:** Cmd + -

Working with Data in a Range

A **range** is a selection of cells in a spreadsheet, often written in the form **A1:B10**, where A1 is the upper-left cell and B10 is the lower-right cell of the range.

Entering Data into Cells

- **Manual Data Entry:**

- Click on any cell (e.g., A1) and begin typing. Press **Enter** to confirm the data entry and move to the cell below, or press **Tab** to move to the right.

- **Copying and Pasting Data:**
  - To copy data from one range to another, select the source range, press **Ctrl + C** (Windows) or **Cmd + C** (Mac), then select the destination range and press **Ctrl + V** (Windows) or **Cmd + V** (Mac).
  - You can also right-click and choose **Copy** or **Paste**.
- **AutoFill:**
  - For repeated patterns, select the initial cells with data, then click and drag the fill handle (a small square at the bottom-right corner of the selected range) to autofill adjacent cells.

Editing Data in a Range

- **Modify Existing Data:**
  - Click on the cell you want to edit and type the new data. Press **Enter** or **Tab** to confirm the change.
- **Fill Series:**
  - If you want to generate a series (e.g., numbers, dates), type the first two numbers (e.g., "1" and "2" in cells A1 and A2) and drag the fill handle to continue the pattern.

2. Selecting Data Ranges

A range refers to a selection of adjacent or non-adjacent cells, and selecting ranges allows you to perform actions like copying, formatting, and calculating.

Basic Range Selection

- **Continuous Range:**

    - Click and drag to select a range of adjacent cells. For example, to select from A1 to C10, click on cell A1, drag your mouse to C10, and the range will be highlighted.
- **Non-Adjacent Range:**
    - To select non-contiguous cells, press and hold **Ctrl** (Windows) or **Cmd** (Mac), and click on the individual cells or ranges you want to select.
- **Entire Row/Column:**
    - Click the row number (e.g., 1) or column letter (e.g., A) to select the entire row or column. For multiple rows or columns, click and drag across the row numbers or column letters.

Range Names

- **Named Ranges:**
    - A **Named Range** is a user-defined name assigned to a specific range of cells. This allows you to refer to that range by name rather than its cell reference.
    - To create a named range:
        1. Select the range.
        2. Right-click and choose **Define Name** (Excel) or **Named Range** (Google Sheets).
        3. Enter a descriptive name for the range, such as SalesData or RevenueRange.
    - You can now use the name in formulas, such as =SUM(SalesData).

3. Working with Ranges in Formulas

Spreadsheet formulas often involve working with data ranges to perform calculations, like summing or averaging values, or applying conditional logic.

Basic Formulas with Ranges

- **SUM:** Adds up the numbers in a range.

=SUM(A1:A10)

AVERAGE: Calculates the average of a range of cells.

=AVERAGE(B1:B10)

COUNT: Counts the number of cells in a range that contain numbers.

=COUNT(C1:C10)

VLOOKUP / HLOOKUP:

- **VLOOKUP** is used to search a value in the first column of a range and return a value from another column in the same row.

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Copy code

=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

- **HLOOKUP** does the same but searches in the first row instead of the first column.

Dynamic Ranges in Formulas

- **Using Dynamic Ranges:**
  - Use **Excel Tables** or **Google Sheets Named Ranges** to make ranges dynamic, meaning the range automatically updates when data is added or removed. For instance, a **Table** in Excel automatically adjusts the range used in formulas when new rows or columns are added.

4. Sorting and Filtering Data

Sorting Data

- **Sort by Columns:**
  - **Excel/Google Sheets:**
    1. Select a column or range of data.
    2. In the **Data** tab, click **Sort**.
    3. Choose either **Sort A to Z** (ascending) or **Sort Z to A** (descending).

Filtering Data

- **Basic Filter:**
  - **Excel/Google Sheets:**
    1. Select the range of data or the entire table.
    2. Click **Filter** from the **Data** tab to apply a filter.
    3. Use the dropdown arrows that appear in the column headers to filter based on specific criteria.
- **Advanced Filtering:**

- ◦ **Excel** allows you to set custom criteria to filter data based on conditions like greater than, less than, or specific text matches.

5. Formatting Data in Ranges

Applying Conditional Formatting

- **Excel/Google Sheets:**
  - Conditional formatting allows you to automatically apply formatting (such as colors, fonts, or borders) based on the values in a range.
  - For example, highlight cells with values greater than 100:
    1. Select the range.
    2. Go to **Home → Conditional Formatting → Highlight Cells Rules → Greater Than**.
    3. Enter the value and choose a format.

Custom Number Formatting

- **Number Formats:**
  - You can apply custom number formats, such as currency, percentage, date, and time, using the **Format Cells** dialog.
  - For example, you can format cells to display numbers as currency by selecting the range, right-clicking, and choosing **Format Cells → Currency**.

6. Data Validation in Ranges

Data validation is useful for restricting the type of data that can be entered into a range of cells.

- **Setting Validation Rules:**
    - **Excel/Google Sheets:**
        1. Select the range you want to validate.
        2. Go to the **Data** tab and click **Data Validation**.
        3. Set the validation criteria, such as allowing only whole numbers or dates within a certain range.

Entering data into worksheet :

1. Basic Data Entry

Entering Text:

- Click on the cell where you want to enter the text.
- Start typing. When you're done, press **Enter** to confirm and move down to the next cell, or **Tab** to move to the next cell on the right.

Entering Numbers:

- Click on the cell where you want to enter the number.
- Simply type the number and press **Enter** to move to the next row or **Tab** to move to the next column.

Entering Dates:

- Click on the cell where you want to enter a date.
- Type the date in a recognizable format (e.g., **MM/DD/YYYY** or **DD/MM/YYYY**).

    - Excel and Google Sheets will automatically recognize the format and apply the date format.
    - **Example**: 12/17/2024 will be interpreted as December 17, 2024.

Entering Time:

- Click on the cell where you want to enter a time.
- Type the time in a recognizable format (e.g., **HH:MM AM/PM** or **HH:MM**).
    - **Example**: 10:30 AM or 14:30 will be recognized as 10:30 AM or 2:30 PM.

2. Data Entry Shortcuts

Using the Enter Key:

- Press **Enter** to move to the next cell directly below the current cell. This is useful when entering data in a vertical column.

Using the Tab Key:

- Press **Tab** to move to the next cell to the right. This is useful when entering data horizontally across a row.

Use Arrow Keys for Navigation:

- Use the **Arrow Keys** on your keyboard to move between cells in any direction (up, down, left, right).

Auto-Complete:

- **Excel/Google Sheets** automatically suggests entries based on previous data entries. For example, if you type "January" in a cell and start typing "Feb", Excel/Google Sheets will suggest "February" based on the first letter or pattern you've entered.

3. Copying and Pasting Data

Copying Data:

- **Copy** data by selecting a cell or range of cells, right-clicking, and selecting **Copy**, or using the keyboard shortcut **Ctrl + C** (Windows) or **Cmd + C** (Mac).
- Paste the copied data into a new location by selecting the destination cell, right-clicking, and selecting **Paste**, or using the keyboard shortcut **Ctrl + V** (Windows) or **Cmd + V** (Mac).

Pasting Values Only:

- When pasting data, you can choose to only paste the **values** and not the formulas. This can be done by selecting **Paste Values** from the paste options (right-click menu or ribbon).

Using Drag-and-Drop for Copying:

- Click and hold the small square at the bottom-right corner of a selected cell or range (called the **Fill Handle**). Drag it to copy the data to adjacent cells.

4. Filling Data

AutoFill:

- **AutoFill** allows you to quickly fill a series of numbers, dates, or other data patterns by dragging the fill handle (the small square in the bottom-right corner of a selected cell).
  - **Example:** If you type "1" in a cell and "2" in the cell below it, select both cells, then drag the fill handle to auto-fill the series (3, 4, 5, etc.).

Custom Fill Series:

- **Excel/Google Sheets** can automatically generate various series, such as days of the week or months. For instance, typing "Monday" and dragging the fill handle will auto-complete the days of the week (Monday, Tuesday, Wednesday, etc.).
- To fill dates, type a starting date (e.g., 01/01/2024), then drag the fill handle to fill subsequent dates.

5. Data Validation for Controlled Entry

Data validation allows you to set rules that control the type of data entered into a worksheet. This can help ensure data consistency and avoid errors.

Steps to Apply Data Validation:

1. Select the range of cells where you want to apply validation.
2. Go to the **Data** tab (Excel) or **Data** menu (Google Sheets).
3. Select **Data Validation**.
4. Set rules for the type of data allowed:
   - **Whole numbers** or **Decimals** (e.g., allowing only numbers between 1 and 100).

    - **Text length** (e.g., only allowing text entries with a specific character length).
    - **Date/Time** (e.g., only allowing dates after a specific date).
    - **Drop-down lists** for predefined choices (e.g., a list of valid categories like "Yes" and "No").

6. Special Data Entry Techniques

Entering Formulas:

- Formulas are used to perform calculations in a spreadsheet.
    - Start by typing an equal sign (=) followed by the formula.
    - **Example:** =A1 + B1 will sum the values in cells A1 and B1.
    - Excel and Google Sheets offer functions like **SUM**, **AVERAGE**, **IF**, **VLOOKUP**, etc., that you can use within formulas.

Concatenating Data:

- Concatenation combines text from multiple cells into one cell.
    - **Example:** =A1 & " " & B1 will combine the text in cell A1 and cell B1 with a space in between.
    - You can also use the **CONCATENATE** function: =CONCATENATE(A1, " ", B1).

Inserting Special Characters:

- You can insert special characters like bullet points, copyright symbols, or other symbols.
    - In Excel, go to **Insert → Symbol** to find special characters.

    - In Google Sheets, go to **Insert → Special characters**.

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7. Copying Data Across Multiple Sheets

Referring to Data in Other Sheets:

- If you want to reference data from another sheet within the same workbook:
    - **Excel/Google Sheets**: Type =Sheet2!A1 to reference cell A1 from **Sheet2**.

Copying Data to Another Sheet:

- You can copy a range of data from one sheet to another by copying the data and pasting it in a new sheet, just like copying between cells within the same sheet.

8. AutoCorrect and Spell Check

AutoCorrect:

- **Excel/Google Sheets** automatically corrects common spelling errors (e.g., changing "teh" to "the"). This feature can be customized in the settings.

Spell Check:

- **Excel:** Go to the **Review** tab and click **Spelling** to check the entire worksheet or a specific range.
- **Google Sheets:** Spell check is automatic as you type, with squiggly red lines under errors.

9. Inserting Data from External Sources

Importing Data:

- You can import data from external sources such as CSV files, databases, or other spreadsheets into your worksheet.
  - **Excel:** Go to **Data → Get External Data** or **Get Data**.
  - **Google Sheets:** Go to **File → Import** and choose the source of the data (e.g., CSV, Excel file).

Copy-Paste from External Sources:

- You can also copy data from websites, other documents, or applications and paste it into your worksheet, adjusting for formatting as necessary.

Saving & quitting worksheet :

Saving and quitting a worksheet is an essential part of working with spreadsheets like Microsoft Excel and Google Sheets. It ensures that your work is preserved and allows you to close the document when you're done. Here's a detailed explanation of how to save and quit a worksheet in different scenarios:

1. Saving Your Worksheet

In Microsoft Excel

1. **Saving for the First Time:**
   - **File → Save As**:
     1. Click on **File** in the top-left corner.
     2. Choose **Save As** from the menu.

    3. Choose a location on your computer (e.g., Desktop, Documents) or in a cloud service (like OneDrive).
    4. Name your file and choose the file format (e.g., **.xlsx** for an Excel workbook).
    5. Click **Save**.
2. **Saving Existing Workbooks (Subsequent Saves):**
    - **File → Save** or **Ctrl + S** (Windows) / **Cmd + S** (Mac):
        1. After the first time you save the file, simply pressing **Ctrl + S** (or **Cmd + S** on Mac) will save any changes you've made since the last save.
        2. Alternatively, you can click **File → Save** in the top menu.
3. **Saving to a Different Location or Format:**
    - **File → Save As**:
        1. If you want to save the workbook under a different name or in a different location, go to **File → Save As**.
        2. Choose the desired location (cloud storage, local folder) and give it a new name if necessary.
        3. You can also change the file format if needed (for example, saving it as a **.csv** or **.pdf**).

In Google Sheets

1. **Automatic Saving:**
    - Google Sheets automatically saves your work in real-time. As long as you are connected to the internet, all changes are saved to **Google Drive** without needing to click any save buttons.

- You will see **"All changes saved in Drive"** appear at the top of the screen, indicating that your work is being automatically saved.

2. **Manual Download (Save a Copy Locally):**
   - If you want to save a copy of your Google Sheets document locally, follow these steps:
     1. Click **File** in the top menu.
     2. Select **Download**.
     3. Choose the file format you want, such as **Microsoft Excel (.xlsx)**, **PDF Document (.pdf)**, or **Comma Separated Values (.csv)**.
     4. The file will then be downloaded to your local system.

2. Quitting (Closing) the Worksheet

In Microsoft Excel

1. **Quitting the Program (Closing Excel and All Workbooks):**
   - **File → Close**:
     - To close the currently open workbook, click **File** in the menu, then select **Close**.
     - If you have multiple workbooks open, this will close only the active workbook, not Excel itself.
   - **Click the Close Button** (X):
     - Click the **X** in the top-right corner of the Excel window to close Excel.
     - If you have unsaved changes, you'll be prompted to save before closing.

2. **Closing Without Saving (Optional):**
   - If you don't want to save your changes, click **Don’t Save** when prompted.
3. **Exiting Excel (Windows/Mac):**
   - **Windows**: Click the **X** in the upper-right corner or press **Alt + F4** to exit Excel.
   - **Mac**: Click the **X** in the upper-left corner of the Excel window or press **Cmd + Q** to quit Excel.

In Google Sheets

1. **Closing a Tab or Window:**
   - **Close the Tab**: In Google Sheets, you can simply close the browser tab or window where your worksheet is open. Since Google Sheets automatically saves your data, you don’t need to worry about saving it manually.
   - **Sign Out of Google Account (Optional)**: If you want to sign out of your Google account and stop accessing Google Sheets, click on your profile picture in the top-right corner of the screen, then select **Sign out**.
2. **Using the Browser Close Button (X):**
   - Simply close the browser tab or window where your Google Sheet is open. Any changes will automatically be saved to Google Drive before closing.
3. **Leaving the Document Open in Google Sheets:**
   - If you plan to come back later, just leave the document open. You don't need to explicitly save or quit it, as Google Sheets continuously saves your work in the cloud.

3. Handling Unsaved Changes

In Microsoft Excel:

- If you attempt to close a workbook with unsaved changes, Excel will prompt you with a message asking if you want to **save**, **discard**, or **cancel** the closure:
  - **Save**: Saves the workbook and closes it.
  - **Don't Save**: Closes the workbook without saving changes.
  - **Cancel**: Cancels the close action and keeps the workbook open.

In Google Sheets:

- Since Google Sheets automatically saves changes as you work, you won't be prompted with a "save" warning when closing the tab or document. However, if you're working offline, it's a good idea to ensure your internet connection is restored so all changes are synced to Google Drive.

4. Creating Backup Copies or Version History (Optional)

In Microsoft Excel:

- You can create backups of your workbook or enable **AutoSave** if you're using **OneDrive** or **SharePoint**:
  - **File → Save As** to create a backup.
  - Enable **AutoSave** by selecting **Save As** to a cloud location such as **OneDrive**.

In Google Sheets:

- **Version History**: Google Sheets automatically tracks changes and allows you to view or revert to previous versions:
    1. Go to **File → Version history → See version history**.
    2. You can view all saved versions and restore any of them if needed.

5. Quitting Without Saving (Specific Scenarios)

In Microsoft Excel:

- If you want to exit Excel and discard any unsaved work:
    - Click the **X** or press **Alt + F4** (Windows) or **Cmd + Q** (Mac).
    - When prompted, click **Don't Save** to quit without saving.

In Google Sheets:

- Since Google Sheets automatically saves your changes, you can simply close the tab or sign out of Google Sheets without worrying about losing data.

Opening and moving around in an existing worksheet :

1. Opening an Existing Worksheet

In Microsoft Excel

1. **Opening from File Explorer (Windows):**
    - **File Explorer**: Navigate to the location where your Excel file is stored (e.g., **Documents**, **Desktop**, or a specific folder).
    - Double-click the Excel file (e.g., **.xlsx** or **.xls**).
    - The file will open in Microsoft Excel.
2. **Opening from Excel Application:**

- Open Excel first by clicking on the Excel icon in your Start menu (Windows) or Applications folder (Mac).
- Click **File** in the top menu and then select **Open**.
- Choose a file from the recent list or click **Browse** to locate and open the file from a specific folder.

3. **Opening from OneDrive or Cloud Storage (If Using Office 365):**
   - If your workbook is stored in **OneDrive** or **SharePoint**, open Excel and select **File → Open → OneDrive** (or another cloud storage service).
   - You can also open Excel directly from **OneDrive** or **SharePoint** in your web browser.
4. **Using Search to Open Files:**
   - If you know the name of the file, you can use the **Search** feature in Windows or macOS to find and open your Excel file directly.

In Google Sheets

1. **Opening from Google Drive:**
   - Go to **Google Drive** by navigating to **drive.google.com** in your web browser.
   - Find the file you want to open by browsing or searching for it using the search bar.
   - Double-click the file to open it in Google Sheets.
2. **Opening from Google Sheets Application (Web Version):**
   - If you are already logged into your Google account, go directly to **sheets.google.com**.
   - You'll see a list of recent documents. Click the one you want to open.
3. **Opening from Email or Shared Link:**

- If someone shared the file with you, open the email with the shared link or go to the **Shared with Me** section in Google Drive and open the file from there.

2. Moving Around in the Worksheet

Once your worksheet is open, you need to know how to navigate through it. This includes moving between cells, rows, columns, and sheets.

Basic Navigation

1. **Using Arrow Keys:**
   - Use the **Arrow Keys** on your keyboard to move up, down, left, or right within the worksheet.
   - Each press of the arrow key will move the active cell one step in that direction.
2. **Moving to a Specific Cell:**
   - **In Excel**: Click on any cell to make it the active one. You can also type the cell reference in the **Name Box** (next to the formula bar) to jump to a specific cell (e.g., A1, B10).
   - **In Google Sheets**: Similarly, click on a cell or type the cell reference (e.g., B5) in the box next to the formula bar to move directly to that cell.
3. **Using the Tab Key:**
   - Press the **Tab** key to move to the next cell to the right.
   - To move left, press **Shift + Tab**.
   - This is useful for entering data in a horizontal direction (across a row).
4. **Using Enter Key to Move Down:**
   - Press **Enter** to move the selection down one row.

    - If you want to move up, press **Shift + Enter**.

3. Navigating Large Worksheets

If you are working with large datasets that span across many rows or columns, here are some methods for moving around efficiently.

Scrolling:

- Use the **Scroll Bars** on the right (for vertical scrolling) and the bottom (for horizontal scrolling) of the window to navigate large worksheets.
- You can also use the **mouse wheel** or **trackpad gestures** to scroll up and down or left and right.

Jumping to the Beginning or End of a Worksheet:

- **In Excel**:
    - Press **Ctrl + Home** (Windows) or **Cmd + Home** (Mac) to jump to the beginning of the worksheet (cell A1).
    - Press **Ctrl + End** (Windows) or **Cmd + End** (Mac) to jump to the last used cell in the worksheet (bottom-right of the data).
-

Navigating Between Sheets (Tabs) in a Workbook:

- **In Excel**:
    - At the bottom of the workbook window, you'll see tabs for each sheet (e.g., **Sheet1**, **Sheet2**).
    - Click on a tab to switch to that sheet.
    - Alternatively, use **Ctrl + Page Up** (Windows) or **Cmd + Fn + Up Arrow** (Mac) to move to the previous sheet.

- Use **Ctrl + Page Down** (Windows) or **Cmd + Fn + Down Arrow** (Mac) to move to the next sheet.

4. Finding Specific Data in the Worksheet

If you're trying to find specific information in a large worksheet, use the **Find and Replace** feature.

In Microsoft Excel:

1. **Find:**
   - Press **Ctrl + F** (Windows) or **Cmd + F** (Mac) to open the **Find** dialog box.
   - Type the text or number you're looking for.
   - Press **Find Next** to locate the next instance.
2. **Replace:**
   - Press **Ctrl + H** (Windows) or **Cmd + H** (Mac) to open the **Find and Replace** dialog box.
   - You can replace specific text or values with something else.

5. Using the Go To Function (Advanced Navigation)

In Microsoft Excel:

1. **Go To Function**: If you want to quickly navigate to a specific cell or range of cells:
   - Press **F5** or **Ctrl + G** (Windows) or **Cmd + G** (Mac) to open the **Go To** dialog box.
   - Type the cell reference (e.g., B10) or range (e.g., A1:C5) to go directly to that location.

Toolbars and menu, keyboard shortcuts :

Toolbars

Toolbars are typically a row or panel of icons or buttons that offer quick access to frequently used features.

1. **Purpose**:
   - Provide shortcuts to essential functions.
   - Enhance productivity by reducing the need for navigating through menus.

1.9Toolbars and menu, keyboard shortcuts

Toolbars, menus, and keyboard shortcuts are essential components of user interfaces that improve navigation and efficiency. Here's a general overview:

Toolbars

Toolbars are typically a row or panel of icons or buttons that offer quick access to frequently used features.

1. **Purpose**:
   - Provide shortcuts to essential functions.
   - Enhance productivity by reducing the need for navigating through menus.

Menus

Menus are a hierarchical list of commands and options.

1. **Types**:
    - **Main Menu**: Found at the top of the screen or window (e.g., File, Edit, View).
    - **Context Menu**: Right-click menu offering quick actions.
    - **Dropdown Menu**: Expands when an item is selected.
2. **Common Menu Items**:
    - **File**: New, Open, Save, Print, Exit.
    - **Edit**: Undo, Redo, Cut, Copy, Paste.
    - **View**: Zoom In/Out, Full Screen, Toolbars.

Excel Shortcuts:

General

F1= Open help

F4= Repeat last action

F7= Run spellcheck

F11= Insert chart in new sheet

Alt= Snap to grid

Alt + F1= Insert embedded chat

Alt + F8= Open macro dialog box

Alt + F11= Open VBA Editor

Alt + '= Open Modify Cell Style

Alt + = Activate filter

Alt + C = Clear select filter

Alt Space= Display control menu

Ctrl + Z= Undo last action

Ctrl + Y= Redo last action

Ctrl + C= Copy selection

Ctrl + X= Cut selection Ctrl + V= Paste content

Ctrl + F= Display Find and Replace(Find Tab)

Ctrl + H= Display Find and Replace(Replace Tab)

General

Ctrl + T= Create table

Ctrl + A= Select table

Ctrl + D= Duplicate object

Ctrl + 6= Hide or show objects

Ctrl + Alt + V= Display the Paste

Ctrl + Shift + L= Toggle Autofilter

Ctrl + Space= Select table column

Ctrl + Shift + F4= Find previous match (after initial Find)

Shift + F4= Find next match (after initial Find)

Shift + FShift + F11= Insert new worksheet

Ctrl + PgDn= Go to next worksheet

Ctrl + PgUp= Go to previous worksheet

Alt + O, H R= Rename current worksheet

Alt + E, L= Delete current worksheet

Alt + E, M= Move current worksheet

ScrLk= Toggle scroll lock

Ctrl + Shift + F1= Toggle full screen

Ctrl + P= Print

Alt + P, R S= Set print area

Alt + P, R C= Clear print area

Alt + R, P S= Protect sheet

7= Open Thesaurus

Shift + F10= Show right-click menu

Shift + Space= Select table row

Worksheet

Shift + F11= Insert new worksheet

Ctrl + PgDn= Go to next worksheet

Ctrl + PgUp= Go to previous worksheet

Alt + O, H R= Rename current worksheet

Alt + E, L= Delete current worksheet

Alt + E, M= Move current worksheet

ScrLk= Toggle scroll lock

Ctrl + Shift + F1= Toggle full screen

Ctrl + P= Print

Alt + P, R S= Set print area

Alt + P, R C= Clear print area

Alt + R, P S= Protect sheet

Workbook

Ctrl + N= Create new workbook

Ctrl + O= Open workbook

Ctrl + S= Save workbook

F12= Save as

Ctrl + Tab= Go to next workbook

Ctrl + Shift + Tab= Go to previous workbook

Ctrl + F9= Minimize

Ctrl + F10= Maximize

Alt + R, P W= Protect workbook

Ctrl + F4= Close current workbook

Alt + F4= Close Excel

Ctrl + F1= Expand or collapse ribbon

Alt = Activate access keys

= Move through Ribbon tabs and groups

Space OR Center= Open selected control

Enter= Confirm control change

F1= Get help on selected control

Drag and Drop

Drag= Drag and cut

Ctrl + Drag= Drag and copy

Shift + Drag= Drag and insert

Ctrl + Shift + Drag= Drag and insert copy

Alt + Drag= Drag to worksheet

Ctrl + Drag= Drag to duplicate worksheet

Navigation

→ = Move one cell right

← = Move one cell left

↑ = Move one cell up

↓ = Move one cell down

Alt + PgUp= Move one screen left

Alt + PgDn= Move one screen right

PgUp= Move one screen up

PgDn= Move one screen down

Ctrl + → = Move to right edge of data

Ctrl + ← = Move to left edge of data

Ctrl + ↑ = Move to top edge of data

Ctrl + ↓ = Move to bottom edge of data

Home= Move to beginning of row

Ctrl + End= Move to last cell in worksheet that contains data

Ctrl + Home= Move to rst cell in worksheet

End= Turn End mode on

Ctrl + End= Move to last cell in worksheet that contains data

1. Single Workbook Operations

Copying Sheets

- **Excel**:
  - Right-click the sheet tab > Select **"Move or Copy"** > Check **"Create a copy"** > Choose the destination.
  - Drag the sheet tab while holding the **Ctrl key** (Windows) or **Option key** (Mac) to create a copy.

Renaming Sheets

- **Excel/Google Sheets**:
  - Double-click the sheet tab to edit the name.
  - Alternatively, right-click the tab > Select **"Rename"**.

Moving Sheets

- **Excel**:
  - Drag the sheet tab to a new position within the same workbook.
  - Right-click the tab > Select **"Move or Copy"** to move it to another workbook.

Adding Sheets

- **Excel/Google Sheets**:
  - Click the **"+" button** at the bottom near the tabs.

- Alternatively, right-click any sheet tab > Select **"Insert"** > Choose the type of sheet.

Deleting Sheets

- **Excel/Google Sheets**:
  - Right-click the tab > Select **"Delete"**.
  - Confirm deletion (irreversible in some cases).

2. Working with Multiple Workbooks

Copying Sheets Between Workbooks

- **Excel**:
  - Open both workbooks.
  - Right-click the tab in the source workbook > Select **"Move or Copy"** > Choose the target workbook > Check **"Create a copy"**.
  - Drag the sheet directly between workbooks (Ctrl/Option key to copy).

Renaming Workbooks

- **Excel**:
  - Save the workbook with a new name using **File > Save As**.

Moving Sheets Between Workbooks

- **Excel**:
  - Use **Move or Copy** to transfer sheets from one workbook to another without duplicating.

3. Copying Entries and Moving Between Workbooks

Copying Entries

- **Between Sheets in the Same Workbook**:
  - Select the data > Use **Ctrl+C** (Copy) > Navigate to the target sheet > Use **Ctrl+V** (Paste).
- **Between Different Workbooks**:
  - Open both workbooks.
  - Copy the data from the source workbook and paste it into the target workbook.

Moving Entries

- **Cut and paste:**
  - Select the data > Use **Ctrl+X** (Cut) > Navigate to the target location > Use **Ctrl+V** (Paste).

Switching Between Workbooks

- **Excel**:
  - Use **Ctrl+Tab** (Windows) or **Cmd+~** (Mac) to toggle between open workbooks.

Different Views of Work sheets:

1. Normal View (Default View)

- **Purpose:** Used for standard data entry and editing.
- **Features**:
  - Displays rows, columns, and gridlines.
  - Allows access to formulas, charts, and formatting options.

2. **Page Layout View**

- **Purpose:** Helps visualize how the worksheet will look when printed.
- **Features:**
  - Displays margins, headers, footers, and page breaks.
  - Shows the sheet with a ruler and page boundaries.

3. **Page Break Preview**

- **Purpose:** Allows users to adjust page breaks for printing.
- **Features:**
  - Displays page numbers and breaks as dotted or solid blue lines.
  - Enables dragging page boundaries to adjust them manually.

How to Access:

- **Excel:** Click on the "View" tab > Select "Page Break Preview".
- **Google Sheets:** Use the Print Preview and adjust the print range.

4. **Custom Views**

- **Purpose:** Lets users save specific views or configurations of a worksheet.
- **Features:**
  - Saves display settings such as hidden rows/columns or filters.
  - Quickly switch between views without affecting the data.
- **Use Case:** Useful for working with multiple filtered data sets or display configurations.

How to Access:

- **Excel:**
  - Click on the "View" tab > Select "Custom Views".
  - Add a new view with your preferred settings.

5. Full-Screen View

- Purpose: Maximizes screen space by hiding the ribbon and menus.
- Features:
    - Displays only the worksheet content.
    - Ribbon and menu bar are hidden.
- Use Case: Ideal for presenting or working with large datasets.

How to Access:

- **Excel:** Press Ctrl+Shift+F1 (Windows) or Command+Option+R (Mac).

6. Protected View

- Purpose: Opens files in a read-only mode to protect against potentially unsafe content.
- **Features:**
    - Disables editing until the user enables it.
    - Often used when files are downloaded from untrusted sources.

7. Split View (Split Panes)

- **Purpose:** Allows viewing and working on different sections of the same worksheet simultaneously.
- **Features:**
    - Splits the worksheet horizontally, vertically, or both.
    - Each section scrolls independently.
- **Use Case:** Useful for large datasets to compare rows and columns.

How to Access:

- **Excel:**
  - Go to the "View" tab > Select "Split".

8. Freeze Panes

- **Purpose:** Locks rows or columns to remain visible during scrolling.
- **Features:**
  - Freezes the top row, left column, or custom sections.
- **Use Case:** Helpful for keeping headers or key references visible.

How to Access:

- **Excel:**
  - Go to the "View" tab > Select "Freeze Panes".

9. Zoom View

- **Purpose:** Adjusts the display size of the worksheet without affecting the actual data.
- **Features:**
  - Allows zooming in and out for better visibility.
- Use Case: Useful for focusing on detailed sections or for overall visualization.

How to Access:

- **Excel:**
  - Use the zoom slider in the bottom-right corner.
-

10. Gridline and Headings Toggle

- **Purpose:** Adjusts the visibility of gridlines and headings for a cleaner view.

- **Features:**
  - Gridlines: Show or hide the lines separating cells.
  - Headings: Show or hide row numbers and column letters.
- **Use Case:** Useful for screenshots or presentations.

How to Access:

- Excel: View > Show > Toggle Gridlines or Headings.

1. Column Freezing

Purpose: Locks specific rows or columns in place to keep them visible while scrolling through a worksheet.

How to Freeze Columns/Rows:

- **Excel**:
  1. Go to the **View** tab.
  2. Select **Freeze Panes**.
  3. Options:
     - **Freeze Top Row**: Locks the top row.
     - **Freeze First Column**: Locks the first column.
     - **Freeze Panes**: Freezes rows and columns up to the selected cell.

2. Labels

Purpose: Refers to headers or titles that help identify the type of data in rows or columns.

Key Features:

- **Excel**:

    - Use the first row as column headers or labels.
    - Apply formatting to distinguish labels (e.g., bold, center-aligned, colored cells).

Freezing Labels:

- Freeze the header row to keep labels visible while scrolling (see "Column Freezing").

Using Labels in Functions:

- Some software allows named ranges or headers to be used in formulas for easier referencing.
-

3. Hiding Columns/Rows

Purpose: Temporarily remove specific rows or columns from view without deleting data.

How to Hide:

1. Select the column(s) or row(s).
2. Right-click > Choose **Hide**.

How to Unhide:

Highlight the surrounding columns/rows.

- Right-click > Choose **Unhide**.

4. Splitting

Purpose: Splits the worksheet view into separate panes, allowing users to view and work on different parts of the same worksheet simultaneously.

How to Split:

- Go to the **View** tab.
- Click **Split** to divide the window into horizontal, vertical, or both panes.
- Adjust split lines by dragging them.

Removing Split Panes:

Go to the **View** tab.

1. Click **Split** again to remove the split.

5. Unhiding and Grouping

Hiding Entire Worksheets:

- Right-click a sheet tab > Select **Hide**.
- To unhide: Right-click any sheet tab > Select **Unhide** and choose the sheet.

Here's a comprehensive guide on **Column Freezing, Labels, Hiding, Splitting, and Related Features** in spreadsheet tools like Microsoft Excel and Google Sheets:

1. Column Freezing

Purpose: Locks specific rows or columns in place to keep them visible while scrolling through a worksheet.

How to Freeze Columns/Rows:

1. Go to the **View** tab.
2. Select **Freeze Panes**.
3. Options:
   - **Freeze Top Row**: Locks the top row.
   - **Freeze First Column**: Locks the first column.
   - **Freeze Panes**: Freezes rows and columns up to the selected cell.

- **Google Sheets**:
  1. Go to **View > Freeze**.
  2. Options:

        - **No rows/columns**: Unfreezes everything.
        - **Up to row/column X**: Freezes up to the specified row or column.

2. Labels

Purpose: Refers to headers or titles that help identify the type of data in rows or columns.

Key Features:

- **Excel**:
    - Use the first row as column headers or labels.
    - Apply formatting to distinguish labels (e.g., bold, center-aligned, colored cells).
- **Google Sheets**:
    - Same as Excel, with options to format the first row distinctly.

Freezing Labels:

- Freeze the header row to keep labels visible while scrolling (see "Column Freezing").

Using Labels in Functions:

- Some software allows named ranges or headers to be used in formulas for easier referencing.

3. Hiding Columns/Rows

Purpose: Temporarily remove specific rows or columns from view without deleting data.

How to Hide:

- **Excel**:
    1. Select the column(s) or row(s).
    2. Right-click > Choose **Hide**.
- **Google Sheets**:
    1. Select the column(s) or row(s).
    2. Right-click > Choose **Hide column** or **Hide row**.

How to Unhide:

- **Excel**:
    1. Highlight the surrounding columns/rows.
    2. Right-click > Choose **Unhide**.
- **Google Sheets**:
    1. Look for small arrows where the hidden rows/columns are located.
    2. Click the arrow to unhide.

4. Splitting

Purpose: Splits the worksheet view into separate panes, allowing users to view and work on different parts of the same worksheet simultaneously.

How to Split:

- Go to the **View** tab.
- Click **Split** to divide the window into horizontal, vertical, or both panes.
- Adjust split lines by dragging them.

Removing Split Panes:

1. Go to the **View** tab.
2. Click **Split** again to remove the split.

5. Unhiding and Grouping

Hiding Entire Worksheets:

- Right-click a sheet tab > Select **Hide**.
- To unhide: Right-click any sheet tab > Select **Unhide** and choose the sheet.

Grouping Rows or Columns:

Purpose: Collapse or expand multiple rows/columns for a cleaner view.

1. Select the rows/columns you want to group.
2. Go to the **Data** tab > Click **Group**.
3. Use the **+/- buttons** to collapse or expand groups.

6. Column Width and Row Height Adjustments

Purpose: Resize rows and columns for better visibility or formatting.

How to Adjust:

- **Manually**:
  - Drag the border between column letters or row numbers.
- **Automatically (Auto-fit)**:
  - **Excel**: Double-click the border between headers (e.g., between column letters A and B).

. Data Filtering

Purpose: Temporarily hide rows that don’t meet specified criteria.

How to Apply Filters:

1. Go to the **Data** tab > Click **Filter**.
2. Dropdown arrows appear on each column header for filtering.

Using Different Features with Data and Text

1.1 Text Features

1. **Text to Columns**:
   - **Purpose**: Splits text in one column into multiple columns based on delimiters like commas, spaces, or tabs.
   - **Excel**:
     1. Select the column with the text.

2. Go to the **Data** tab > Click **Text to Columns**.
3. Choose **Delimited** or **Fixed Width** > Select delimiter (e.g., comma, tab).
4. Finish to split the data.

Concatenate (Combine Text):

- **Purpose**: Combines text from multiple cells into one.
- Formula:
  - Excel/Google Sheets: =CONCATENATE(A1, " ", B1)
  - Or use =TEXTJOIN(" ", TRUE, A1, B1) for more flexibility.
- Shortcut: Use the & operator: =A1 & " " & B1.

Change Case (Uppercase, Lowercase, Proper Case):

- **Excel**:
  - Use functions:
    - =UPPER(A1) for uppercase.
    - =LOWER(A1) for lowercase.
    - =PROPER(A1) for proper case.

Remove Unwanted Spaces:

- Use =TRIM(A1) to remove extra spaces from text.

1.2 Data Features

1. **Sorting and Filtering**:
   - **Sort**:

Sort data in ascending/descending order based on one or more columns.

Excel: Go to the **Data** tab > Select **Sort**.

Filter:

- Apply filters to show only relevant data.
- **Excel**: Go to **Data > Filter**.

Conditional Formatting:

- Highlights cells based on specific conditions.
- **Excel/Google Sheets**: Go to **Home > Conditional Formatting**.

Data Validation:

- Restricts data entry to specific types or values.
- **Excel**: Go to **Data > Data Validation**.
- **Google Sheets**: Use **Data > Data Validation**.

Find and Replace:

- Quickly replace text or values in the worksheet.
- Shortcut: **Ctrl + H** (Excel/Google Sheets).

2. Advanced Paste Special Techniques

Purpose:

The Paste Special feature allows you to control how data or formatting is pasted.

Common Paste Special Options:

- **Values Only**:
    - Pastes the values without formulas or formatting.

    - Shortcut: **Ctrl + Alt + V > V** (Excel).
- **Formulas Only**:
    - Pastes only formulas from the copied cells.
- **Formats Only**:
    - Copies cell formatting (e.g., color, font) without data.
- **Transpose**:
    - Converts rows to columns and vice versa.
    - Shortcut in Excel: **Ctrl + Alt + V > E**.
    - Google Sheets: Use **Paste Special > Transpose**.

Advanced Paste Special Use Cases:

1. **Multiply, Add, Subtract, Divide**:
    - **Excel**:
        1. Copy a number (e.g., 2).
        2. Select the range you want to modify.
        3. Use **Ctrl + Alt + V > Multiply/Add/Subtract/Divide** to apply the operation to all selected cells.
2. **Skip Blanks**:
    - Avoids overwriting data with blank cells when pasting.
    - **Excel**: Use **Ctrl + Alt + V > Skip Blanks**.
3. **Linking Data**:
    - Paste data as a linked reference to the original cells.
    - **Excel**: Use **Ctrl + Alt + V > Paste Link**.